

Course Syllabus

CPT_S 451

Introduction to Database Systems

Course Details

Instructor & TA Information

- Instructor: Parteek Kumar
 - Office location: EME B55
 - Email: parteek.kumar@wsu.edu
 - Phone: 509-335-7114
 - Office hours: Tuesday and Thursday, 12:30 PM – 1:30 PM, and by appointment for additional availability. Friday, 4:30 to 5:30 PM (Only on Zoom)
 - Virtual Option: Zoom available for coaching and support
 - Zoom Link: [Join HereLinks to an external site.](#)
 - Meeting ID: 731 933 4661
 - Passcode: 998629
- Teaching Assistant(s): TBD

Course Overview

This course provides students with an introduction to the design of databases and the use of database management systems for

applications. We will cover the design and implementation of traditional relational database systems as well as advanced data management systems. We will discuss the fundamental principles of databases such as the (ER) approach to database design, the relational model, relational design theory, abstract query language such as relational algebra, and programming in SQL. Students will get exposure to how relational database management systems are used to implement a database. We will also cover core database implementation issues including storage and indexing, and transaction management.

Prerequisites: "Cpts 223/233 Advanced Data Structures", certification in major.

Student Learning Outcomes (SLOs)

- To learn how to design effective database schemas that accurately model real-world information semantics.
- To learn how data is represented in relational model and how integrity constraints are expressed and enforced in the relational model to ensure data consistency and accuracy.
- To learn how to pose complex database queries in SQL as well as theoretical query languages.
- Learn about the normal forms and their purposes. Understand the benefits of BCNF and learn the considerations in decomposing relations into appropriate normal forms.
- To understand what is meant by a transaction. To gain an introductory background in concurrency control and recovery.
- To learn about indexing in database systems. Given a query, be able to evaluate which indexes are likely to be most effective

Course Topics

- Basics of DBMS, File-Based System vs. Centralized System, Data Models

- Three-Level Architecture
- Data Independence and How DBMS Accesses Data
- RDBMS Keys and Applying Integrity Constraints
- ER Model Concepts and Conversion to Tables
- Normalization Concepts and Normal Forms
- Storage Mechanisms and Indexing Concepts
- Transaction Management and ACID Properties
- Query Optimization Techniques
- Introduction to Big Data and NoSQL Systems
- SQL Basics: Table Creation, Data Manipulation, SELECT, Filtering, and Aggregation
- SQL: Working with Date and Time
- SQL: Application of Inbuilt Functions
- SQL: Aggregate Functions and Grouping of Data
- SQL: Joining of Tables
- SQL: Indexes and Views

Expectations for Student Effort

For each hour of lecture, students should expect to devote a minimum of two to three hours of work outside of class.

Since this is a 3-credit course, you should plan for approximately 6–9 hours per week beyond scheduled class meetings. This time includes readings, homework assignments, labs, and project work.

Assignments, Assessments, and Grading Policy

What you'll submit and how it's assessed

- Homework: Assigned weekly, with one assignment per week aligned to the topics covered in class.
- Project (semester-long, team): Milestones include reports, code, tests; individual grades may be adjusted via peer evaluation.
- Class Tests: Short online checks for conceptual mastery.
- Midterm & Final Exams: Closed-book assessments of core concepts and applied reasoning.
- Participation: Evaluated through sampled attendance and regular in-class activities/assignments. Active engagement in discussions, group tasks, and class exercises will contribute to this component of the grade.

Submission Standards

Unless specified otherwise, submit a single PDF via Canvas by 11:59 PM on the due date.

Grade Scale

The final course grade will be calculated using the breakdown shown in Table 2. Numeric scores will then be converted into letter grades according to the standard grade scale.

Grading Breakdown

Table 2. Components and Weights

Grading breakdown	
Component	Weight
Class Participation	5%
Class Activities	5%
Homework Assignments	30%

Project	30%
Midterm exam	12%
Final exam	13%
Class Tests	5%

In addition to these components, informal in-class quizzes will be conducted weekly during the last class session using Kahoot. The winner of each quiz will earn extra credit, with a maximum of 2 extra credit points available over the course of the semester. Awarding and calculation of bonus points will be at the instructor's discretion at the end of the semester.

Rounding Policy

Final course percentages will be calculated to two decimal places and then rounded once to the nearest whole number when converting to letter grades:

- Percentages ending in .50 or higher will round up (e.g., 89.50% → 90%).
- Percentages ending in .49 or lower will round down (e.g., 89.49% → 89%).

No additional rounding, bumping, or discretionary adjustments will be applied beyond this rule.

Letter Grade Scale

Grading scale breakdown

Score Range	Grade
≥93	A
90–92	A–
86–89	B+
83–85	B

80–82	B–
76–79	C+
73–75	C
70–72	C–
66–69	D+
64–65	D
60–63	D–
<60	F

End Semester Exam Date and Time

CPT_S 451-02 (5015)	Intro to Database Syst (Lecture)	5/5/2026, Tuesday	10:30AM - 12:30PM	The Spark G0010	
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Communication

All course communication will be managed through Canvas. Announcements, assignments, lecture materials, and other learning resources will be posted there regularly.

Class-related Q&A will take place on the Canvas Discussion Board, which also serves as the portal for project deliverables, homework submissions, and grading. For questions about lectures, assignments, or project milestones, please post to the Discussion Board instead of emailing, so that the entire class can benefit from shared responses and discussions. Be sure to subscribe to all forums so you don't miss important updates or extended discussions.

For emergencies, accommodations, or extension requests, you should email the instructor directly. Please include a clear subject line (e.g., CptS 451 – Extension Request) to ensure your email is noticed quickly.

The instructor and TA will generally respond to Discussion Board posts and emails within 24–48 hours on weekdays.

Late Submission Policy

All homework and project milestones are due on the posted deadlines to keep the class on schedule.

- If you request in advance: Email the instructor at least 24 hours before the deadline to ask for an extension. Approved extensions will typically carry a 1% deduction per day late. In cases of emergencies or documented circumstances, this penalty may be waived.
- If you miss the deadline without approval: Late work will be accepted with a 5% deduction per day, up to a maximum of 30%. Submissions more than one week late are normally not accepted, unless explicitly approved by the instructor under special circumstances.

Homework and Project

Homework

- Homework will be assigned weekly, with one assignment per week aligned to class topics.
- Assignments will be announced in class and posted on Canvas with due dates.
- Homework is designed to reinforce lecture content and typically requires steady weekly effort rather than last-minute work.
- Students are strongly encouraged to use instructor and TA office hours for clarification and support.

Project Policy

The semester-long project will be organized into three sprints, each approximately one month long.

Sprint Deliverables

For each sprint, your team must submit:

- GitHub repository (with commits showing contributions from all members)
- Kanban board (updated regularly to reflect task progress)
- Video submission briefly describing each member's contributions and showing demos/sample code
- Sprint report (written summary of progress, design decisions, and testing)

Peer Evaluation & Grading

Each sprint grade will be based on two components:

- Team Deliverable Grade (1/3): Quality and timeliness of the team's submitted work.
- Peer Evaluation Grade (2/3): Individual contributions as evaluated by peers.

This weighting ensures that individual accountability is emphasized more heavily than the group result.

Example:

- Team receives 10 points for Sprint 1.
- A student receives 8/10 in peer evaluation.
- Final Sprint 1 grade for that student = $(10 \times 1/3) + (8 \times 2/3) = 3.3 + 5.3 = 8.6$.

Final Project Demo

At the end of the semester, all teams will deliver a live demo and final project report. Peer evaluation will again be used to adjust individual scores.

Participation & Attendance Policy

Active participation and regular attendance are essential for success in this course.

- Attendance & Credit
 - Attendance will be sampled randomly throughout the semester.
 - Attending $\geq 85\%$ of sampled lectures earns the full participation credit (5% of the course grade). To provide flexibility, the lowest 15% of attendance sessions will be dropped when calculating your attendance score. This allows students to miss up to 7 sessions for valid reasons (e.g., illness, official university events) without penalty.
 - Bonus points may be awarded for attendance above 85%.
- In-Class Activities & Engagement (5%)
 - Participation will also be evaluated through regular in-class activities and assignments, such as discussions, group tasks, and class exercises.
 - Active engagement in these activities contributes directly to the participation grade.
 - The lowest 15% of class activity scores will be dropped, giving students some flexibility while encouraging active engagement.

- Submission Deadline: All class activities are due by 11:59 PM on the same day the activity is introduced in class.
- Excused Absences
 - If you anticipate a justifiable absence (e.g., university activity, religious holiday, military service, medical appointment), email the instructor at least 12 hours in advance when possible.
 - For emergencies or illness, notify the instructor as soon as feasible.
 - Documentation from medical personnel is not required, but a short written explanation is expected. Activity supervisors should provide documentation for university events.
- Making Up Missed Material
 - If you miss class for an excused reason, you are responsible for reviewing notes from a classmate and consulting posted materials. Use office hours if you need clarification.
- Professional Conduct
 - Silence or mute personal electronics (non-disruptive use for class purposes is allowed).
 - Arrive on time and remain for the full session.
 - Contribute respectfully to class discussions and group activities.

Make-Up Policy

- Exams: Make-ups allowed with prior approval or documented emergency; complete within 1 week of the missed date.
- Labs/Assignments: Arrange in advance or within 1 week after an emergency, subject to instructor approval.

Academic Integrity Statement

You are responsible for reading WSU's [Academic Integrity Policy](#)

[Links to an external site.](#)

, which is based on [Washington State law](#)

[Links to an external site.](#)

In this course, academic integrity means completing your own work honestly, responsibly, and in alignment with course expectations. Examples of violations include (but are not limited to):

- Submitting copied or plagiarized work (from classmates, books, the internet, GitHub, or AI tools without permission).
- Misrepresenting group project contributions.
- Using unauthorized materials, notes, or devices on assignments, exams, or quizzes.
- Falsifying data, test results, or project deliverables.

If you cheat in your work in this class you will:

- Fail the assignment in question. Serious or repeated violations may result in a failing grade for the entire course.

- Be reported to the [Center for Community Standards](#)[Links to an external site.](#)
- Retain the right to appeal my decision.
- Not be allowed to drop or withdraw from the course until the appeals process is complete.

If you are unsure about what is or is not allowed in this course, ask me before submitting the work.

If you wish to contest a decision regarding academic integrity, you must submit an appeal using the [Academic Integrity Appeal Form](#) within 21 calendar days of the decision.

Course AI Use Policy

Generative AI tools (e.g., ChatGPT, GitHub Copilot) may be used in this course only under the specific conditions below. This policy is an extension of WSU's [Academic Integrity Policy](#) and violations will be treated as academic integrity violations.

1. Allowed with Disclosure

- Brainstorming ideas, outlining project activities, or refining drafts.
- Generating non-graded practice questions or debugging suggestions.
- Using AI tools (e.g., Copilot) for coding support, provided contributions are explicitly documented.

2. Allowed When Explicitly Permitted

- Some assignments will clearly state that AI use is allowed (e.g., to improve design or testing work).

- If an assignment does not mention AI use, assume it is not allowed.

3. Required Disclosure

- Any AI use must be acknowledged directly in the submission. Examples:
 - “Used ChatGPT to brainstorm test case names.”
 - “GitHub Copilot suggested initial loop structure.”
- Sprint submissions must include a dedicated section describing how AI was (or was not) used during project work.

4. Not Allowed

- Submitting AI-generated code, text, or designs as your own without disclosure.
- Using AI to complete graded work when not explicitly permitted.
- Any undisclosed use of AI will be considered a violation of academic integrity.

5. Academic Integrity Reminder

All AI use must support—never replace—your own learning. Undisclosed or unauthorized AI use will result in sanctions under the WSU Academic Integrity Policy.

Accessibility and Accommodations

WSU is committed to providing equal access to learning opportunities for all students. Reasonable accommodations are available for those with documented disabilities or chronic medical/psychological conditions.

If you need accommodations, please contact your campus Access Center/Access Services as early as possible (ideally within the first two weeks of the semester). Once accommodations are approved, it is your responsibility to meet with the instructor (via email, Zoom, or in person) to discuss how they will be implemented in this class.

Refer for further details: [Disability Accommodations – Students | Global Campus Proctoring Services | Washington State University](#)

[Links to an external site.](#)

Campus Contacts:

- [Pullman, Global Campus, and Everett Access Center](#)[Links to an external site.](#)

509-335-3417, Access.Center@wsu.edu

Accommodation for Religious Observances or Activities

WSU reasonably accommodates absences for reasons of faith or conscience. Provide written notification within the first two weeks of the course including the specific dates. Approved accommodations will not adversely impact grades.

For further details, please refer: [Religious Accommodations | Access at WSU | Washington State University](#)

[Links to an external site.](#)

Safety and Emergency Information

- Campus Safety Plan — safetyplan.wsu.edu
- WSU Emergency Management — oem.wsu.edu

- WSU Alert System — alert.wsu.edu
Follow “Alert, Assess, Act” and “Run, Hide, Fight” as appropriate. Enable WSU Alerts in myWSU.

COVID-19 Policy

Students are expected to abide by current university health directives. Any temporary modality changes or accommodations will follow university guidance and be communicated via Canvas.

University Syllabus (Required Statement)

“Students are responsible for reading and understanding all university-wide policies and resources pertaining to all courses (for instance: accommodations, care resources, policies on discrimination or harassment), which can be found in the University Syllabus.”

Link: syllabus.wsu.edu/university-syllabus

Instructor's Note

Portions of this syllabus were refined using AI tools to improve clarity and formatting. All course policies, requirements, and content are authored and approved by the instructor.